

Model Fitting

Math 742

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Mean Squared Error

There are several metrics for determining the quality of our model fit. Probably the most common method is the Mean Squared Error (MSE).

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$$

In this equation, y_i represents the true response and $\hat{f}(x_i)$ represents the response estimated by our model.

The MSE is computed from the training data and is used as a metric for assessing model fit. Generally, we are more concerned with the accuracy of the model in terms of predicting a new unknown response. But what if we don't have any test data? You might think that in that case you use the model which has the smallest training MSE. This approach can often lead to overfitting (more later on this).

The following simulation illustrates the bias/variance tradeoff of the MSE.

- The data is from a 4th degree polynomial with with noise (black).
- A linear model is fit (orange)
- 2 spline fits are also show, more degrees of freedom (i.e. more flexible) wigglier fit.
- The right panel displays the MSE of the training and test sets.
- The training error decreases monotonically, reflecting improved fit to the observed data.
- Test error initially decreases as bias is reduced and then increases due to growing variance.
- Squares represent the training and test MSEs for the three fits shown in the left-hand panel.
- Irreducible error cannot be eliminated. It arises from inherent randomness in the data generation process such as measurement noise and unobserved factors.

Left: Data simulated from f , shown in black. Three estimates of f are shown: the linear regression line (orange curve), and two smoothing spline fits (blue and green curves).

Right: Training MSE (grey curve), test MSE (red curve), and minimum possible test MSE over all methods (dashed line).

In an earlier lecture I showed that

$$MSE = \mathbb{V}(\hat{f}) + (\text{Bias})^2$$

Reducing the bias increased the variance and vice versa.

The total MSE actually has the 3rd component (irreducible error) in it as well.

$$MSE = \mathbb{V}(\hat{f}) + (\text{Bias})^2 + \mathbb{V}(\varepsilon)$$

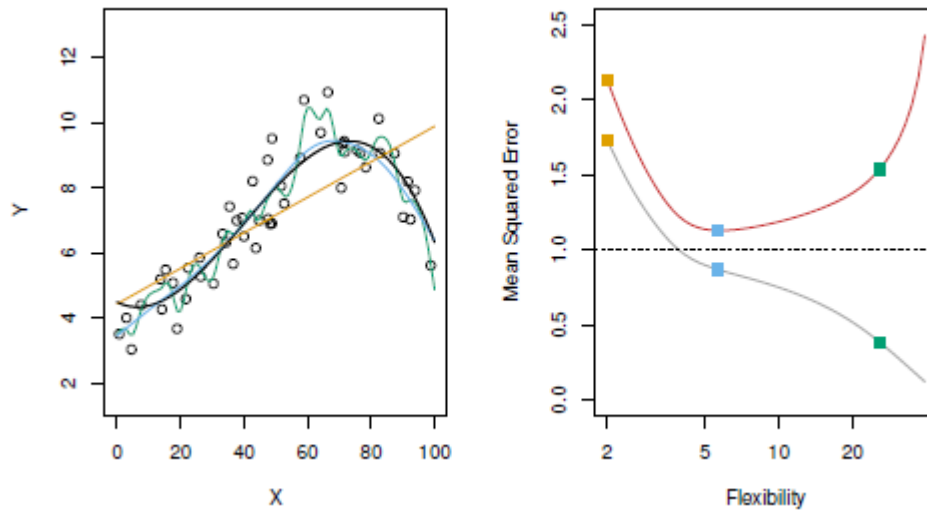


Figure 1: Model Accuracy

Model Selection vs. Model Assessment

Why the distinction matters.

One of the most common and serious mistakes in statistical modeling is using the same data to both choose a model and evaluate its performance. This leads to optimistic bias: the model appears to perform better than it truly will on new data.

To avoid this, we must clearly separate two tasks:

- Model selection: Which model should we use?
- Model assessment: How well does the chosen model perform on new data?

Failing to distinguish these tasks undermines nearly every result that follows.

Model Selection

Definition: Model selection is the process of choosing among a set of candidate models or tuning parameters.

Examples:

1. Choosing polynomial degree
2. Selecting spline degrees of freedom
3. Choosing predictors in regression
4. Choosing the regularization parameter λ in ridge or lasso
5. Choosing between competing model classes (linear vs nonlinear)

There are some tools that you can use to decide which model fits best.

1. R^2 - coefficient of determination.

In order to calculate R^2 you perform analysis of variance on your data:

It's not too hard to show:

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

In words, this equation can be expressed as:

$$\text{Total Sum of Squares} = \text{Model Sum of Squares} + \text{Residual Sum of Squares}$$

Definition:

$$\begin{aligned} R^2 &= \frac{\text{Model Sum of Squares}}{\text{Total Sum of Squares}} = 1 - \frac{\text{Residual Sum of Squares}}{\text{Total Sum of Squares}} \\ &= \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \end{aligned}$$

The coefficient of determination gives you an indication of how much of the total variability is explained by the model.

Another related metric is the **adjusted** R^2 . Adjusted R^2 provides a penalty for having too many terms in the model.

2. Akaike Information Criterion (AIC)

AIC is rooted in information theory → it specifically aims to estimate the relative amount of information lost when a model approximates the true (but unknown) data generating process.

The AIC uses the **Kullback-Leibler (KL) divergence** which is a metric to decide how far a probability density estimate differs from the true density.

$$\text{AIC} = -2\ell(\theta) + 2k$$

where,

ℓ is the maximized value of the log-likelihood function

k is the number of estimated parameters in the model including the intercept

The term $-2\ell(\theta)$ rewards better fit (lower value means better likelihood), while $+2k$ penalizes complexity to avoid overfitting (more parameters make the model more flexible but risk fitting noise rather than signal).

AIC is commonly used in regression, time series (e.g., ARIMA), ecology, psychology, and machine learning for tasks like feature selection or comparing parametric models.

Relationship to KL divergence

- KL divergence measures how far a model distribution is from the true data-generating distribution. It is theoretical because we do not know the true distribution.
- – The expected KL divergence can be approximated (up to an additive constant) by $-2 \log L(\hat{\theta}) + 2k$, which balances model fit and complexity
- $\text{AIC} = -2 \log L(\hat{\theta} \mid \mathbf{y}) + 2k$

AIC is a practical way to choose the model that is expected to have the smallest information loss (i.e., smallest KL divergence) when used to describe new data from the same source. In practice: AIC favors models that predict new data well.

3. Bayesian Inference Criteria (BIC)

BIC is derived from Bayesian principles, specifically as an asymptotic approximation to the log marginal likelihood (also called the model evidence or integrated likelihood) under certain conditions.

In Bayesian statistics, the full way to compare models is to compute the posterior probability of each model given the data:

$$P(\text{Model} \mid \text{Data}) \propto P(\text{Data} \mid \text{Model}) \times P(\text{Model})$$

Under standard regularity conditions, as $n \rightarrow \infty$ the log marginal likelihood can be approximated as,

$$\log(P(\text{Model} \mid \text{Data})) = \ell(\theta) - (k/2) \ln(n) + \text{other terms that don't depend on the model}$$

So BIC is essentially a large-sample approximation to -2 times the log Bayes factor (or to comparing posterior model probabilities when assuming equal prior probabilities on models and using a specific unit information prior).

Comparison between AIC and BIC

Aspect	AIC	BIC
Penalty term	$+2k$	$+k \ln(n)$
Goal	Minimize expected Kullback-Leibler divergence (best predictive model)	Approximate posterior probability of the model being true (best chance the model is the “true” one)
Penalty Strength	Milder — grows linearly with k	Stronger, grows with $\ln(n)$, so heavier for large n
Consistency	Not consistent	Consistent
Best use	Prediction, approximates model well	Model identification when // you believe one model is exactly true
Behavior with large n	Tends to select more complex models	Strongly prefers simpler models

We have a couple of lectures devoted to CV and Penalized likelihood methods, so I'll just briefly outline them here.

4. Cross Validation (CV)

The basic CV procedure, withholds a certain portion of your data as a test set. You fit the model with most of the data and then test your model on the part that was withheld.

5. Penalized likelihood (ridge,lasso)

Ridge: Minimize $\text{RSS} + \lambda \|\beta\|_2 = \text{RSS} + \lambda \sum \beta_j^2$. $(X^T X + \lambda I) X^T y$, closed form solution.

Lasso: Minimize $\text{RSS} + \lambda \|\beta\|_1 = \text{RSS} + \lambda \sum |\beta_j|$, no closed form solution (usually).

Comparison of model Metric:

```
set.seed(1)

# -----
# 1) Simulate data
# -----
n <- 300

# Gender as a factor
gender <- factor(sample(c("F","M"), n, replace = TRUE))

# SAT (roughly 400-1600 scale, centered near ~1100)
sat <- round(pmin(pmax(rnorm(n, mean = 1100, sd = 180), 400), 1600))

# HS GPA (0-4 scale, mildly correlated with SAT)
hs_gpa <- 2.8 + 0.0012*(sat - 1100) + rnorm(n, 0, 0.35)
hs_gpa <- pmin(pmax(hs_gpa, 0), 4)

# True model for first-year college GPA (0-4)
# (You can tweak these coefficients to match your narrative.)
beta0 <- 0.8
beta_sat <- 0.0008
beta_hs <- 0.55
beta_male <- -0.12 # M slightly lower on average (holding others fixed)

# Create outcome with noise
eps <- rnorm(n, mean = 0, sd = 0.35)

gpa1 <- beta0 +
  beta_sat * sat +
  beta_hs * hs_gpa +
  beta_male * (gender == "M") +
  eps

gpa1 <- pmin(pmax(gpa1, 0), 4)

dat <- data.frame(gpa1, sat, hs_gpa, gender)

# -----
# 2) Fit candidate models
# -----
m1 <- lm(gpa1 ~ sat + hs_gpa + gender, data = dat) # full
m2 <- lm(gpa1 ~ sat + hs_gpa, data = dat) # drop gender
m3 <- lm(gpa1 ~ hs_gpa + gender, data = dat) # drop sat
m4 <- lm(gpa1 ~ sat + gender, data = dat) # drop hs_gpa
m5 <- lm(gpa1 ~ sat + hs_gpa + sat:gender, data = dat) # add interaction (optional)

models <- list(
  "Full: sat + hs_gpa + gender" = m1,
  "No gender: sat + hs_gpa" = m2,
  "No sat: hs_gpa + gender" = m3,
  "No hs_gpa: sat + gender" = m4,
  "Add sat×gender interaction" = m5
```

```

)

# -----
# 3) Build summary table
# -----
model_table <- do.call(rbind, lapply(names(models), function(name) {
  fit <- models[[name]]
  s <- summary(fit)
  data.frame(
    Model = name,
    R2 = s$r.squared,
    Adj_R2 = s$adj.r.squared,
    AIC = AIC(fit),
    BIC = BIC(fit),
    row.names = NULL
  )
}))

# Nice formatting + rank by BIC (you can change to AIC or Adj_R2)
model_table <- model_table[order(model_table$BIC), ]
model_table$R2 <- round(model_table$R2, 4)
model_table$Adj_R2 <- round(model_table$Adj_R2, 4)
model_table$AIC <- round(model_table$AIC, 1)
model_table$BIC <- round(model_table$BIC, 1)

model_table

```

```

##           Model      R2 Adj_R2   AIC   BIC
## 2   No gender: sat + hs_gpa 0.5158 0.5125 225.2 240.0
## 1 Full: sat + hs_gpa + gender 0.5163 0.5114 226.9 245.4
## 5 Add sat*gender interaction 0.5163 0.5114 226.9 245.4
## 3   No sat: hs_gpa + gender 0.4377 0.4339 270.0 284.9
## 4   No hs_gpa: sat + gender 0.3373 0.3328 319.4 334.2

```

- R^2 always weakly increases when you add predictors. Adjusted R^2 , AIC, and BIC penalize complexity.
- BIC penalizes complexity more strongly than AIC, so it tends to pick simpler models as n grows.
- When comparing models fit to the same data, the one with the smaller AIC or BIC is preferred.

All of these criteria rely on the likelihood in some way, which raises a deeper question:

What role should the likelihood play in statistical inference?

Likelihood Principle

We've seen that R^2 always increases (or stays the same) when adding predictors, so it really doesn't help with selection. This makes R^2 unsuitable for model selection without adjustment. Adjusted R^2 adds a small penalty, so it's better, but still ad-hoc and tied to Gaussian linear models. BIC (and AIC) penalize more systematically using the log-likelihood + complexity term. Both AIC and BIC are built directly on the likelihood of the data under the model.

The likelihood principle (Birnbbaum 1962, building on ideas from Fisher and others) says:

Once we fix a statistical model (i.e., a family of probability distributions indexed by parameters θ), all the information the data provide about θ is contained in the likelihood function $L(\theta) = \mathbb{P}(\mathbf{x}|\theta)$.

Two different experiments (or two different datasets) that happen to produce proportional likelihood functions ($L_1(\theta) \propto L_2(\theta)$) as functions of θ should lead us to draw exactly the same conclusions about θ — even if the experiments look very different in terms of stopping rules, sampling schemes, etc.

Anything else (the full sampling distribution, how the data were collected, optional stopping, etc.) is irrelevant for inference about θ once the likelihood is fixed.

Example: Suppose you want to estimate the probability θ of heads on a coin.

- Experiment A: Flip the coin until you get 3 heads. At some point, you observe exactly 3 heads in a row, so the experiment stops.
- Experiment B: Flip the coin exactly 3 times. You observe HHH.

For experiment B, $L_B = \theta^3$. For experiment A, the likelihood is proportional to θ^3 , since all additional terms do not depend on θ and can be absorbed into the proportionality constant.

According to the likelihood principle, the two experiments provide exactly the same evidence about θ .

The two experiments have the same likelihood because, once we condition on observing three heads, all other aspects of the experiment affect the probability only by a constant that does not depend on θ .

Many frequentist procedures (p-values, confidence intervals) would give different answers because they depend on the sampling procedure / stopping rule. The likelihood principle says that is inappropriate, only the likelihood matters.

Birnbaum's Likelihood Principle

All information in the data about the parameter θ is contained in the likelihood function

$$L(\theta | x) = p(x | \theta).$$

Key Implication: If two experiments produce proportional likelihood functions, then they must lead to identical statistical inference about θ .

Interpretation:

- Inference should depend only on the observed data through $L(\theta | x)$.
- Sampling plans, stopping rules, and unobserved outcomes are irrelevant once data are observed.

Connection to Model Selection:

- Methods based purely on likelihood (MLE, Bayesian inference) respect the likelihood principle.
- Criteria such as AIC and BIC use the likelihood but add penalties, so they are not purely likelihood-based.

Connection to AIC/BIC

The likelihood principle tells us we should take the likelihood seriously as the carrier of information from data, it makes sense to build model comparison around how well different models make the observed data probable (i.e., their likelihood). AIC and BIC are both penalized likelihood criteria:

They start from the maximized log-likelihood (goodness-of-fit term), then subtract a penalty that grows with model complexity (number of parameters k)

$$\text{AIC} \approx -2\ell(\theta) + 2k$$

(asymptotic justification: approximates out-of-sample predictive performance / Kullback-Leibler divergence to truth)

$$\text{BIC} = -2\ell(\hat{\theta}) + k \ln n$$

(Bayesian justification: approximates the marginal likelihood / posterior probability of the model under certain priors. It's more aggressive at preferring simpler models as n grows)

AIC and BIC are built from the likelihood, but they are not purely likelihood-based because they include penalty terms that depend on model complexity. They are motivated by likelihood-based reasoning, but they do not strictly satisfy the likelihood principle.

Pure maximum likelihood follows the likelihood principle strictly but tends to overfit (no penalty).

- Adjusted R^2 heuristic penalty, Gaussian-world only.
- AIC follows likelihood principle + aims for good prediction (less penalty, keeps more complex models).
- BIC follows likelihood principle + aims for finding the true model (stronger penalty, prefers parsimony, especially in large n).
- The likelihood principle is the reason most modern model selection (GLMs, mixed models, time series, Bayesian model comparison, etc.) is likelihood-centric rather than, purely cross-validation-centric or purely geometric (like R^2).

Summary

- Likelihood principle: all information about θ is in $L(\theta \mid \text{data})$
- Explains why likelihood drives modern model fitting
- AIC/BIC = penalized likelihood (fit + complexity tradeoff)
- AIC $\rightarrow$ prediction
- BIC $\rightarrow$ model identification (parsimony)